SMART INDIA HACKATHON 2026

DossierGraph

Evidence, not opinions: measured AI pre-scrutiny for AICTE approval dossiers

Problem Statement ID	SIH26261
Problem Statement Title	AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
Theme	[Theme as listed on the SIH portal]
PS Category	Software
Team ID	[Team ID]
Team Name	[Team Name]

DossierGraph: whole-dossier AI pre-scrutiny for AICTE approvals



Proposed solution

An AI pre-scrutiny engine that checks an institution's whole approval dossier: each document against its issuer, against APH 2024-27, and against every other document. Officers get evidence-linked findings; they make every decision.

- **Smart intake:** classify + OCR 8 document types (English + Hindi)
- **Verify first:** DSC, QR and DigiLocker checks before any forensics
- **Dossier graph:** land deed ↔ building plan ↔ OC ↔ fire certificate ↔ declared area ↔ faculty
- **Rules as code:** 20 APH 2024-27 rules, each citing its clause

Example flag: fire certificate address ≠ building-plan address (APH Cl. 6.17(i)), evidence crops attached

How it addresses the problem

- **Self-disclosed papers trusted at entry →** inconsistencies flagged at submission
- **Repetitive manual checks →** triaged queue; evidence + clause on each flag
- **Hard to confirm with issuers →** 6 clear statuses, unresolved cases routed to manual review

Innovation & uniqueness

- **Dossier-level cross-consistency,** not single-document KYC checks
- **Accuracy proven** on an open, labelled forgery benchmark
- **Rules track APH changes;** AI advises, the officer decides

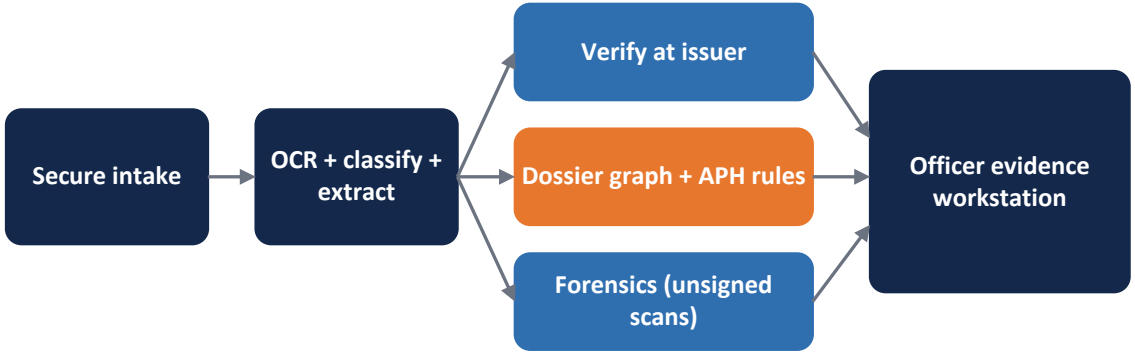
Technical approach

Technologies to be used

Frontend	Next.js · Tailwind
Backend	FastAPI · Celery + Redis
Data	PostgreSQL · MinIO (S3)
AI / OCR	PaddleOCR (EN + Hindi) · OpenCV · doc classifier
Verification	pyHanko (DSC) · QR decode · DigiLocker adapter
Trust layer	Keycloak · OPA · Vault · ClamAV

All open-source, all on-prem. No applicant data is sent to third-party AI APIs. Rules live as versioned YAML, so they can be updated when the APH changes.

Architecture



Methodology

1. Scan, sanitise and encrypt every upload
2. OCR + extract facts with page coordinates
3. Verify signed docs at source; forensics only on scans
4. Build dossier graph; run versioned APH rules
5. Officer reviews evidence; adverse outcomes need a 2nd officer

Feasibility and viability

Technical
Proven open-source parts; no custom model training needed

Operational
Runs before the Scrutiny Committee; APH committee flow unchanged

Economic
Open-source and on-prem; scale by adding worker nodes

Legal & privacy
Aligned to DPDP Act 2023 and MeitY AI guidelines

Potential challenges & risks	Strategies for overcoming them
Poor-quality or regional-language scans	Indic OCR models; low-confidence fields go to manual review
Sophisticated forgeries evade forensics	Verify at issuer first; forensics is multi-signal and advisory
False flags add officer workload	Thresholds tuned on the benchmark; false flags per genuine application tracked
No public APIs for fire NOCs or land records	Pluggable adapters, spec-faithful mocks, 'Source unavailable' status
Sensitive institutional and personal data	Encryption, role-based access, tamper-evident audit log, on-prem AI



How we'll prove it works: an open synthetic benchmark of 200 institutions, ~2,000 documents and 8 labelled tamper types. We report precision and recall per tamper type and false flags per genuine application. No unmeasured accuracy claims.

Impact and benefits

Impact on the target audience

AICTE scrutiny officers
Triaged queue; every flag shows evidence and its APH clause

Institutions
Pre-check before submission means fewer deficiency rounds

Expert Visit Committees
A 'verify on site' list of only what could not be confirmed

Students and public
Approved institutions' claims are better verified

Today vs. with DossierGraph

Today	With DossierGraph
Documents trusted on self-disclosure	Cross-checked against issuers and each other at submission
Deficiencies fixed up to 6 months after EoA	Fixed before submission via pre-check
Findings vary by reviewer	Same versioned APH rules for every application
Inspectors re-check everything	EVC targets only unresolved items

Social
Fewer students at institutions with false claims

Economic
Less officer and inspection time per application

Operational
Earlier deficiency detection; faster processing

Governance
Tamper-evident record of every rule and decision

Research and references

Regulatory

- AICTE Approval Process Handbook 2024-25 to 2026-27: Cl. 2.3.5, 2.4, 6.14, 6.17, 6.18, 7.4; Annex. 3, 5, 14 cdnbbsr.s3waas.gov.in
- AICTE Grant of Approval guidelines 2024-27 indiankanoon.org
- DPDP Act 2023 and DPDP Rules 2025 (notified Nov 2025) ssrana.in
- MeitY India AI Governance Guidelines (Nov 2025) en.vikaspedia.in

Integration

- DigiLocker Requester Entity Locker API Specification, NeGD (Nov 2024) entity.digilocker.gov.in
- pyHanko: PDF digital-signature validation pyhanko.readthedocs.io

Research and prior work

- Grobler et al., A Technique for the Detection of PDF Tampering or Forgery, arXiv:2507.00827 (2025) arxiv.org
- SIH 2024 AICTE PSs SIH1729 (AI approval portal) and SIH1730 (AI inspection) driems.ac.in

How we compare

Capability	KYC tools	Typical SIH entry	DossierGraph
Verify at issuer	✓	Partial	✓
Whole-dossier cross-checks	✗	Partial	✓
APH rules with clauses	✗	Partial	✓
Measured accuracy, open data	Partial	✗	✓
On-prem, DPDP-aligned	Partial	Partial	✓

KYC tools: e.g. IDfy, HyperVerge (document tamper checks for KYC and lending).
Typical SIH entry: based on public SIH26261 repositories, Oct 2026.